

VENKATA VINESH KUMAR REDDY ATLURI

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EDUCATION

École Centrale de Lyon

Lyon Centrale Digital Lab 2026–2027, Diplôme d'établissement

Écully, France

Sep 2026 – Aug 2027

- Full-time team projects (3 to 5 students) building prototypes for partner companies, coached by Centrale Lyon faculty and company professionals. Concludes with a six-month industrial internship from February 2027.

Mahindra University

B.Tech in Computer Science and Engineering, CGPA 7.96/10

Hyderabad, India

Aug 2023 – 2027 (expected)

- Final year completed on exchange at École Centrale de Lyon.
- Relevant coursework: Deep Neural Networks, Reinforcement Learning, Computational Finance with Applications, Applied Time Series Analysis.

TECHNICAL SKILLS

Languages: Python (primary), C++, SQL, TypeScript, MATLAB

ML and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, Statsmodels, LSTM/GRU, time-series forecasting, reinforcement learning (Q-Learning, SARSA)

Data and Systems: NumPy, Pandas, FastAPI, Next.js, React, Supabase, PostgreSQL, Docker, Git

Mathematics: probability, statistics, optimization, stochastic calculus, time series analysis

PROJECTS

Veltrix, Market Analytics and Forecasting Terminal | *TypeScript, Next.js, Supabase, Python* | [repo](#)

Dec 2025 – Present

- Built a signal engine that exposes every indicator vote behind a BUY or SELL call with its weight (RSI, MACD, Bollinger, EMA/SMA trend, volume), summed and normalised to $[-1, +1]$ and thresholded at ± 0.3 , so each recommendation is auditable rather than opaque.
- Validated forecasts walk-forward across 1,500 out-of-sample calls and measured a 50.2% directional hit-rate ($z = 0.16$), reporting the absence of edge rather than a tuned backtest figure. Benchmarked ridge regression and gradient-boosted trees against that baseline and removed both after they failed to beat it at 123 to 443 times the inference cost.
- Replaced an EWMA volatility estimate with GARCH(1,1), raising 95% prediction-interval coverage from 91.7% to 94.75% against nominal.

Asset Price Prediction Platform | *PyTorch, Statsmodels, FastAPI, React* | [repo](#)

Jan 2026 – Present

- Placed a stacked LSTM and an ARIMA baseline on the same series so the trade-off between them is measured rather than assumed.
- Applied index-aligned windowing to prevent look-ahead leakage, and persisted scaler state alongside the model so inference reproduces training-time normalisation.
- Built a Pandas pipeline computing 14-day rolling SMA, EMA, RSI and volatility features, and a FastAPI backend serving multi-step forecasts with intervals to a React dashboard.

Weather Time-Series Forecasting | *PyTorch, Statsmodels* | [repo](#)

Jun 2025 – Jul 2025

- Benchmarked a stacked two-layer LSTM (64 units, dropout 0.2) against ARIMA and SARIMA statistical baselines.
- Decomposed the series into trend, seasonality and residual components, tuning hyperparameters against residual stationarity checks. Runs on a reproducible synthetic series, so results replicate without a dataset download.

Reinforcement Learning Lab | *NumPy, Matplotlib* | [repo](#)

Feb 2026 – May 2026

- Implemented Q-Learning and SARSA with the temporal-difference updates written by hand on a custom Gym-style GridWorld, comparing on-policy and off-policy convergence under ϵ -greedy decay from 1.0 to 0.05.

CERTIFICATIONS AND LANGUAGES

Certification: Machine Learning Specialization, DeepLearning.AI and Stanford Online (Andrew Ng), Aug 2026. Credential ID JOWXFIWH9QN2.

Languages: English (fluent), Telugu (native), Hindi (conversational), French (A1, in progress)

LEADERSHIP AND ACTIVITIES

Mathematics Club, Mahindra University | *Head, promoted from Logistics Head*

Aug 2024 – May 2026

- Logistics Head (Aug 2024 – Apr 2025), then Head (Aug 2025 – May 2026). Coordinated logistics and resource allocation for university mathematics modelling competitions and academic events.

TEDx Mahindra University | *On-Ground Operations Lead*

Jan 2024 – May 2024

- Ran on-ground operations, venue setup and speaker logistics for an event with over 500 attendees.